

5 (a) State Kirchhoff's first law.

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 [1]

(b) Fig. 5.1 shows a circuit containing a thermistor T that has a negative temperature coefficient.

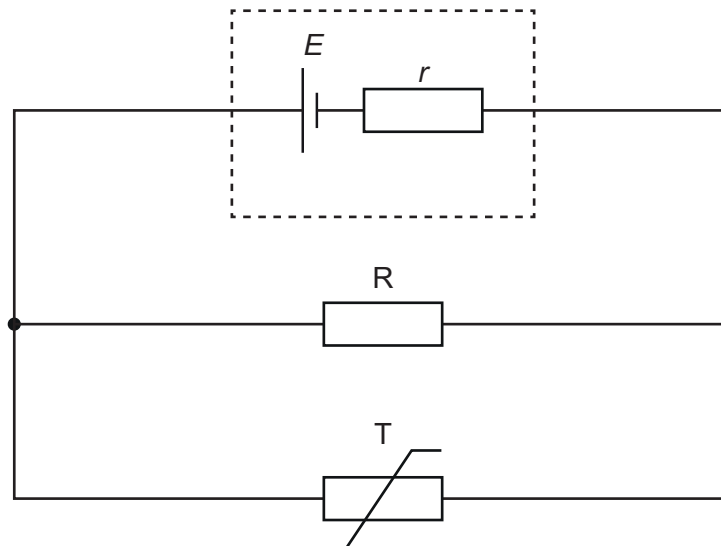


Fig. 5.1

(i) The thermistor has resistance R_0 at a temperature of 0°C .

On Fig. 5.2, sketch a possible variation of the resistance of the thermistor with temperature between 0°C and 100°C .

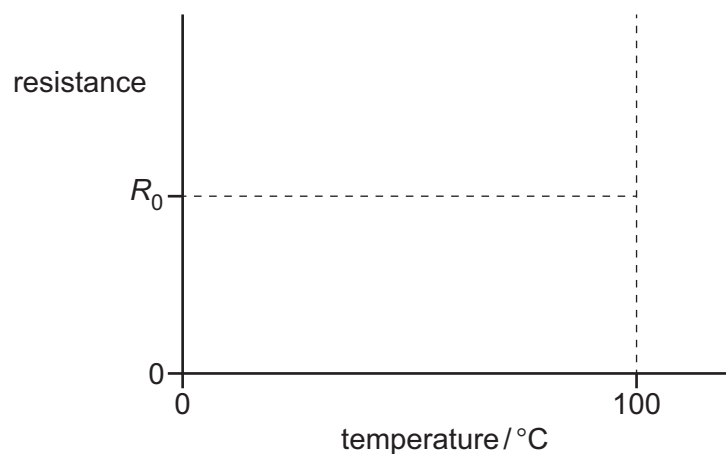


Fig. 5.2

[2]

